

Education

- 2024 – Now **Hong Kong Polytechnic University**, Ph.D. in Computing, *GPA: 4.1/4.3*.
Advisor: Prof. **Wenqi Fan** and Prof. **Qing Li**
- 2021 – 2024 **Sichuan University**, M.E. in Computer Science and Technology, *GPA: 3.76/4*.
Advisor: Prof. **Min Zhu**, Lab: Vision Computing Lab
- 2017 – 2021 **Sichuan University**, B.E. in Computer Science and Technology, *GPA: 3.77/4*.
Under the Wu Yuzhang Honors Program

Research Interests

AI for Science, Multimodal Protein Modeling, Large Language Models, Diffusion Models

Publications & Preprints (First Author)

- 2026 **Yi Zhou**, Haohao Qu, Yunqing Liu, Shanru Lin, Le Song, Wenqi Fan. "HD-Prot: A Protein Language Model for Joint Sequence-Structure Modeling with Continuous Structure Tokens". *International Conference on Knowledge Discovery and Data Mining (KDD)*, 2026. (**CCF-A**)
- 2025 **Yi Zhou**, Wenqi Fan, Shijie Wang, Yuyao Yan, Hui Liu, Qian Zhao, Le Song, Qing Li. "Computational Protein Science in the Era of Large Language Models (LLMs)". *ACM Computing Surveys (CSUR)*, Under Review.
- 2024 **Yi Zhou**, Xian Guan, Meixuan Wu, Chengzhou Ouyang, Min Zhu. "Timely-MDA: A Benchmark for Generalizable MiRNA-Disease Association Prediction". *International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024. (**CCF-B**)
- 2022 **Yi Zhou**, Xinyi Wang, Lin Yao, Min Zhu. "LDAformer: Predicting LncRNA-Disease Associations based on Topological Feature Extraction and Transformer Encoder". *Briefings in Bioinformatics (BIB)*, 2022. (**JCR-Q1, IF: 13.994, CCF-B**)

Publications & Preprints (Collaboration)

- 2026 Tianzhong Lan, Weili Jiang, Yisong Liu, **Yi Zhou**, Junqi Bai, Si Yong Yeo, Xulei Yang, Min Zhu. "Wavelet-Driven Cross-Domain Consistency for Mixed-Supervised 3D Tumor Segmentation". *European Conference on Computer Vision (ECCV)*, 2026. (**CCF-B**)
- Yunqing Liu, **Yi Zhou**, Jun Xia, Wenqi Fan, Qing Li. "Geometric Flow Matching for Conformation Generation via Manifold Decomposition". *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2026, Under Review.
- 2026 Yunqing Liu, **Yi Zhou**, Wenqi Fan. "Enhancing Molecular Property Predictions by Learning from Bond Modelling and Interactions". *International Conference on Learning Representations (ICLR)*, 2026. (**CCF-A**)
- 2025 Lin Gan, Xinyi Wang, **Yi Zhou**, Min Zhu. "Protein-binding RNA Prediction Based on Integrated Sequence-Structure-Function Pre-training". *IEEE Transactions on Computational Biology and Bioinformatics (TCBB)*, 2025. (**JCR-Q1**)
- Meixuan Wu, Yizhou Yang, **Yi Zhou**, Kai Wang, Junqi Bai, Min Zhu. "PLHGMDA: Pre-trained Language Model and Heterogeneous Graph Neural Network for MiRNA-Disease Association Prediction". *International Conference on Intelligent Computing (ICIC)*, 2025. (**CCF-C, Oral**)

- 2024 Wenwen Gao, Shangsong Liu, **Yi Zhou**, Fengjie Wang, Feng Zhou, Min Zhu. "GBDT4CTRVis: Visual Analytics of Gradient Boosting Decision Tree for Advertisement Click-Through Rate Prediction". *Journal of Visualization (JoV)*, 2024.
- Jiamin Zhu, Meixuan Wu, **Yi Zhou**, Nan Cao, Haotian Zhu, Min Zhu. "Dowsing: A Task-Driven Approach for Multiple-View Visualizations Dynamic Recommendation". *Journal of Visualization (JoV)*, 2024.

Tutorials

- 2026 Yujuan Ding, Liangbo Ning, Ziran Liang, Chun-Hin Chan, **Yi Zhou**, Haohao Qu, Wenqi Fan. "Towards Autonomous Web Interaction: Empowering WebAgents by Large Foundation Models". *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2026. (CCF-C)

Patents

- 2024 Min Zhu, Meixuan Wu, **Yi Zhou**. "A MicroRNA-Disease Association Prediction Method Based on Graph Neural Networks". CN202411841640.5.
- 2023 Min Zhu, Jiamin Zhu, Meixuan Wu, **Yi Zhou**. "A Visual Analytics Method for RNA-Disease Associations Based on Density Relational Graphs". CN202311187212.0.
- 2022 Min Zhu, **Yi Zhou**, Xinyi Wang, Lin Yao. "Method and System for Long Non-coding RNA-Disease Association Prediction Based on Self-Attention Mechanism". CN202210818621.5.
- Min Zhu, Xiyao Li, Chunlin Long, **Yi Zhou**, Xinyi Wang. "A Prediction Method for Chromatin Topological Association Domain Boundary Based on Multimodal Fusion". CN202211477002.0.
- Min Zhu, Jiamin Zhu, Meixuan Wu, **Yi Zhou**, Haotian Zhu. "A Dynamic Visualization Recommendation Method Based on User Tasks". CN202211610962.X.
- Min Zhu, Meixuan Wu, Jiamin Zhu, **Yi Zhou**, Haotian Zhu. "An Analytical Task Perception Method that Integrates Deep Learning Models and Rules". CN202211104004.5.
- 2021 Min Zhu, Fuqiu Chen, Chunlin Long, **Yi Zhou**, Xinyi Wang. "A Visualization Method for Chromatin Hierarchy Analysis Based on Genetic Data". CN202111217034.2.

Teaching & Services

- Fall 2023 - **Teaching Assistant**, Sichuan University.
- Spring 2024 - Data Visualization (311301030) × 2
- Fall 2024 - **Teaching Assistant**, Hong Kong Polytechnic University.
- Now - Introduction to Artificial Intelligence and Data Analytic (COMP1004) × 3
- Computational Thinking and Problem Solving (COMP1010)
- Conference Reviewer.**
- KDD 2026 AI for Sciences Track
- Journal Reviewer.**
- Expert Systems With Applications (ESWA)

Honors & Awards

- 2024 PolyU Research Postgraduate Scholarship (PRPgS)
- 2024 Outstanding Undergraduate Teaching Assistant at Sichuan University
- 2024 Outstanding Graduate of Sichuan University
- 2023 Outstanding Graduate Student of Sichuan University
- 2023 First-class of Excellent Graduate Scholarship by Sichuan University
- 2023 Tencent Scholarship of Sichuan University
- 2021 Certificate of Honor from Wu Yuzhang Honors College

Skills

Programming Python, PyTorch, SQL, Linux commands, Git, LaTeX
Languages Mandarin, English (IELTS: 7.0), Fuzhou dialect